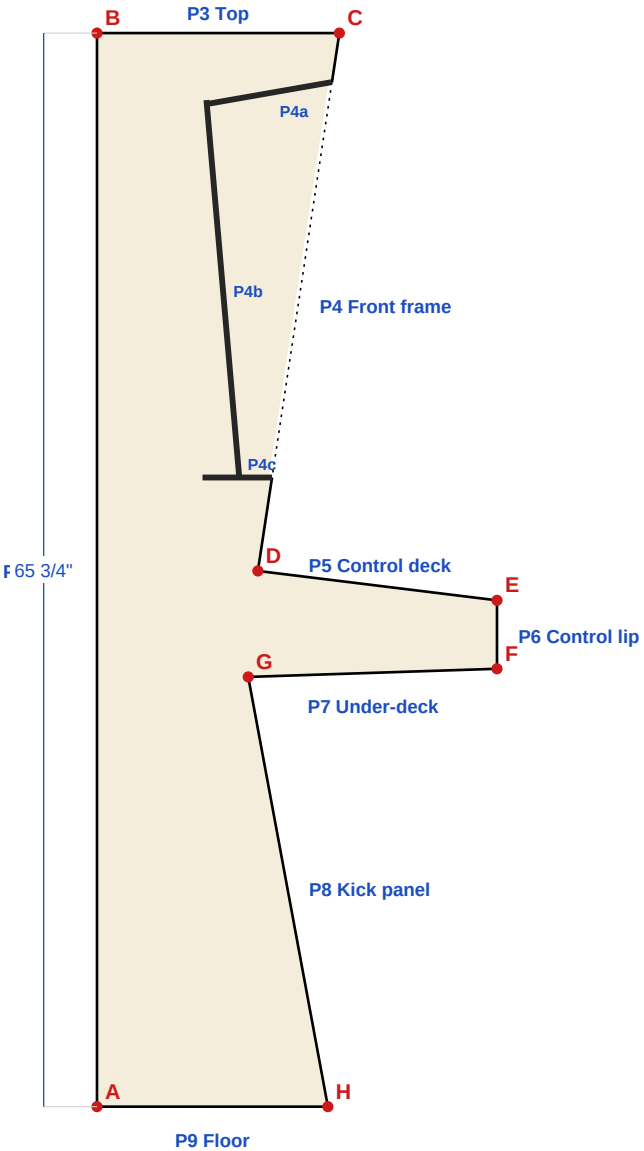
SUPERCADE - cardboard edition

Full size, kid height (8 in lower). Printable plan set, built from the original Supercade CAD files.



P1 SIDE PANEL (x2) - side view, back on the left

AT A GLANCE

Overall height	65 3/4"
Overall depth	24 1/2"
Inside width	29"
Outside width	29" + 2 x side thickness
Control deck height	~31 7/8" (7 deg tilt)
Screen	27" monitor (original), lighter is better
Material	double-wall corrugated cardboard
Fasteners	kids' plastic cardboard screws + glue

WHAT'S IN THIS PACKET

1 This overview
2 Side panel - measure-and-mark drawing
3 Cut list (every part, every size)
4 Cross panels - fold-flange layout
5 Control panel layout + hole schedule
6 Shopping list
7 Assembly steps

1:1 print-and-tape templates in full-size/templates/:
control-panel-1to1.pdf
speaker-cutouts-1to1.pdf
side-panel-1to1-kid.pdf

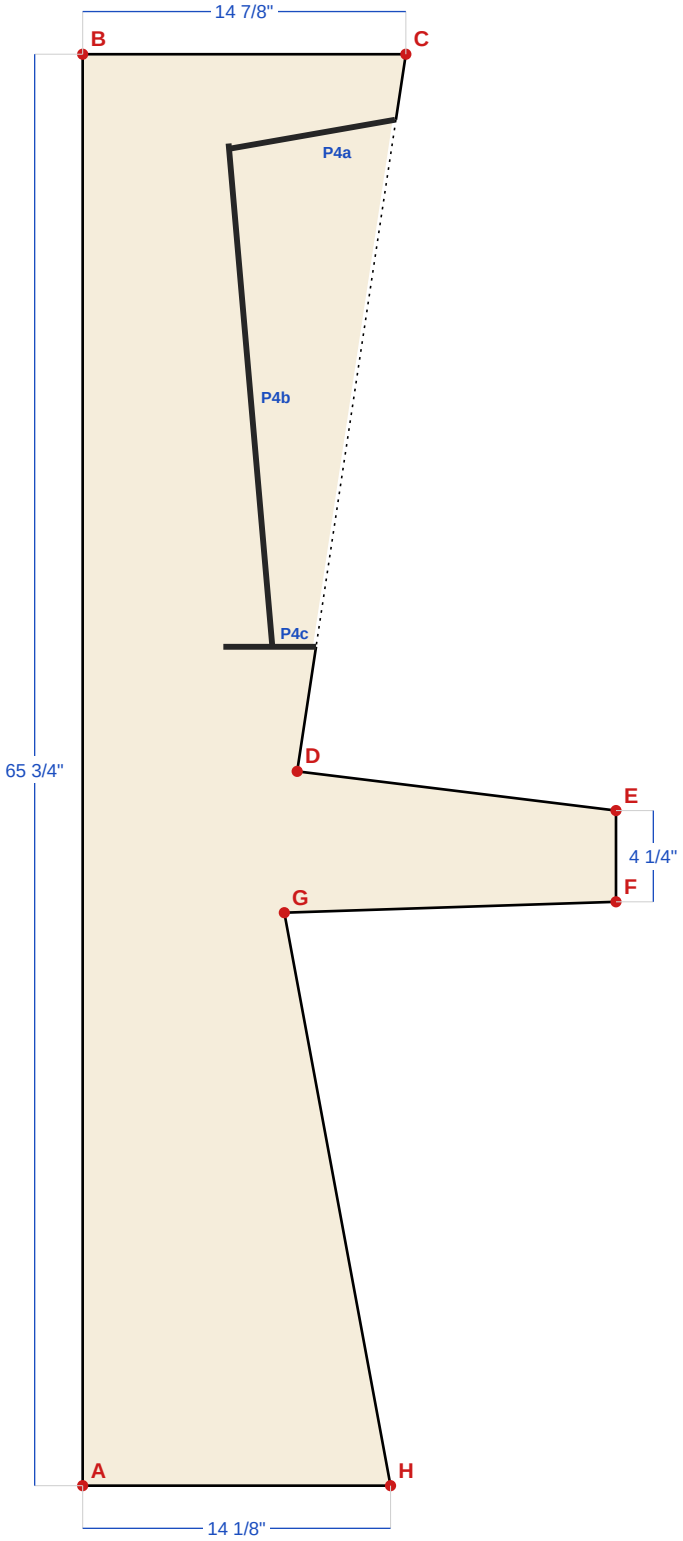
HOW THE CARDBOARD VERSION WORKS

The original is 1/2" plywood held together with wood screws driven into panel edges. Cardboard edges can't hold a screw, so every cross panel (P2-P9) gets a 2" flange folded down on each end. The flange lies flat against the inside of the side panel and a plastic screw + nut goes through both layers, face to face. That's exactly the joint kids' cardboard screw kits are made for.

Side panel shape, panel lengths and control layout are the original's. Only the joinery changes.

P1 Side panel (make 2)

Draw it straight onto the cardboard with a tape measure and a long straightedge. Every edge is a straight line.



MARK THESE POINTS

Lay the sheet flat. Draw the BACK edge first as a long straight line (A to B). Every point is measured UP from the floor end along that line, then OUT square to it (use a set square or a big book).

Pt	Up from floor	Out from back	Is
A	0" (0.00)	0" (0.00)	floor, back corner
B	65 3/4" (65.73)	0" (0.00)	top, back corner
C	65 3/4" (65.73)	14 7/8" (14.84)	top, front corner
D	32 3/4" (32.80)	9 7/8" (9.85)	screen meets deck
E	31" (31.00)	24 1/2" (24.49)	deck, front edge
F	26 3/4" (26.81)	24 1/2" (24.49)	lip, bottom
G	26 1/4" (26.31)	9 1/4" (9.26)	under deck, inner
H	0" (0.00)	14 1/8" (14.14)	floor, front corner

INSIDE LINES (pencil only - where the recess parts go)

S1	61 3/8" (61.39)	6 3/4" (6.74)	P4a speaker shelf, back
S2	62 3/4" (62.72)	14 3/8" (14.34)	P4a speaker shelf, front
M1	38 1/2" (38.52)	8 3/4" (8.71)	P4b screen board, bottom
M2	61 5/8" (61.63)	6 3/4" (6.70)	P4b screen board, top
R1	38 1/2" (38.52)	6 1/2" (6.46)	P4c recess floor, back
R2	38 1/2" (38.52)	10 3/4" (10.72)	P4c recess floor, front

Join the dots in order A-B-C-D-E-F-G-H-A with straight lines. Draw the inside lines S1-S2, M1-M2 and R1-R2 on the INSIDE face: the recess parts' flanges line up on them.

Check your work: D-E should be 14 3/4" long and C-D 33 1/4". If not, a point is off.

Second side: cut the first one out, flip it over, and trace it. The two sides are mirror images, so tracing the flipped panel gives you the correct other hand automatically.

Flutes UP AND DOWN (parallel to A-B): that direction carries the screen.

Stiffness: glue a second layer to each side (flutes crossed), OR glue 3" wide ribs of cardboard on the inside, keeping the 2" strip along every edge free for the flanges. See the screw-length note on page 6 before you pick.

Both sides fit on one 48 x 96 sheet with the second turned end-for-end.

Cut list

Cross panels are 29" between the sides + a 2" fold flange each end = 33" wide blanks.

Part	Name	Qty	Cut size (L x W)	Edge	Notes
P1	Side panel	2	65 3/4" x 24 1/2"	-	Shape on page 2. Mirror pair.
P2	Back	1	65 3/4" x 33"	A-B	Cut a 12" x 20" service door low down (optional).
P3	Top	1	14 7/8" x 33"	B-C	
P4	Front frame	1	33 1/4" x 33"	C-D	ONE continuous flat piece. Cut an opening 27" wide, from 3" to 27 1/2" down from the top (C) edge. The recess sits behind it.
P4a	Speaker shelf	1	7 3/4" x 33"	S1-S2	Recess ceiling. Two 5 1/4" x 4" speaker holes, pointing DOWN at the players.
P4b	Screen board	1	23 1/4" x 33"	M1-M2	Centred window cut to YOUR monitor's visible area (about 23 1/2" x 13 1/4" for a 27"). Leans back 5 degrees; the monitor bolts or straps on behind it.
P4c	Recess floor	1	4 1/4" x 33"	R1-R2	Bottom of the recess, level with the top of the frame's lower lip.
P5	Control deck	1	14 3/4" x 33"	D-E	Laminate 2-3 layers. Tape the 1:1 control-panel template on, centred.
P6	Control lip	1	4 1/4" x 33"	E-F	Front face under the players' wrists.
P7	Under-deck	1	15 1/4" x 33"	F-G	
P8	Kick panel	1	26 3/4" x 33"	G-H	Leg room slope. Optional access door.
P9	Floor	1	14 1/8" x 33"	H-A	Double layer. This carries the ballast.
P10	Screen shelf	1	27" x 33"	-	Folds into a 10" deep x 3" thick box beam (page 4). Sits under the monitor; set its height on the dry-run.
P11	Lower shelf	1	27" x 33"	-	Same box beam, low down. Holds the computer and the ballast.
P12	Deck doubler	2	14 3/4" x 29"	-	Glued under P5 so the controls don't flex. Cut holes through all layers together.
P13	Corner strips	8	2" x 29"	-	Folded into L-angles and glued into inside corners that feel loose.
P14	Screen cover	1	23 1/4" x 29"	-	Optional. Original is 1/4" dark-tint acrylic. Clear PET sheet works.

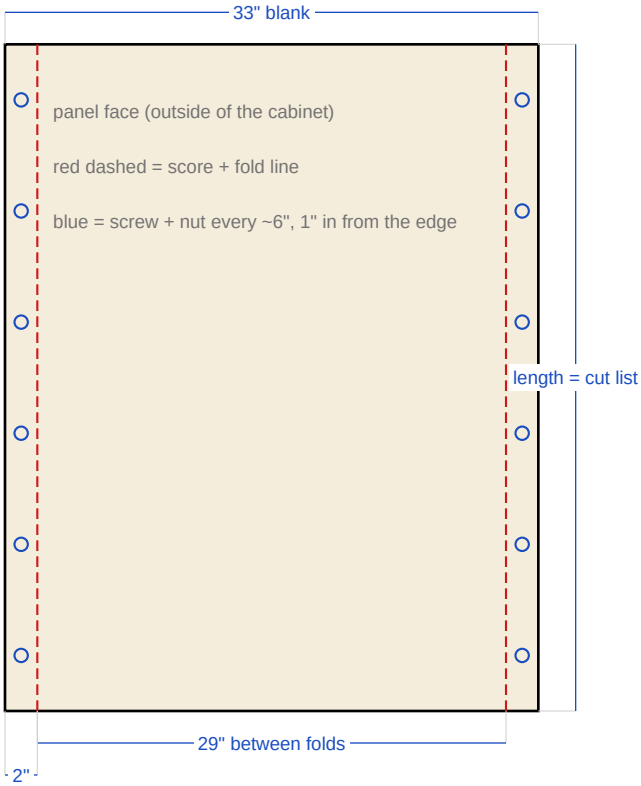
Cardboard: roughly five 48" x 96" double-wall sheets. A refrigerator box gets you the two sides and the back.

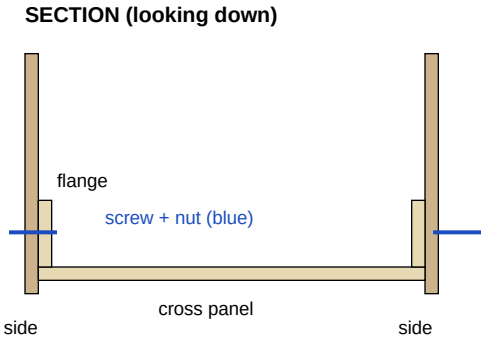
Every size is measured on the CAD edge. Cut cross panels 1/4" long and trim during the dry fit: cardboard is thicker than the pencil line, so corners will need a nibble.

Flutes: on the cross panels, run them ACROSS the 33" width so the flange fold goes along the flutes (cleaner fold, stiffer span).

Cross panels - fold-flange layout

The same pattern for P2-P9. Score, fold 90 degrees, screw to the side panels.





P10 / P11 BOX-BEAM SHELF

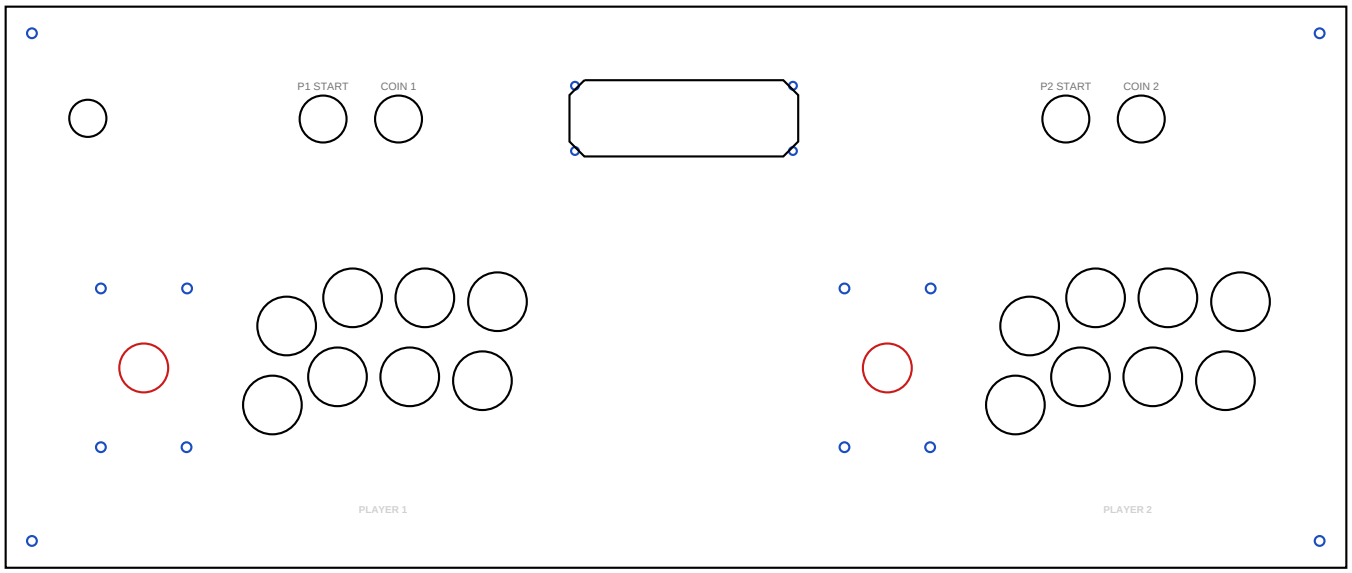


A 27" x 33" blank. Cut away the two grey strips (2" x 17") so the end flanges only remain on the 10" top. Score every red line. Fold the rest under into a closed 10" x 3" tube and glue the tab to the underside of the top. A closed tube is what makes a cardboard shelf stiff enough to hold a monitor. The end flanges screw to the side panels.

Screen shelf: set its height so the monitor sits centred behind the P4b window. Stand the monitor on it with its own stand removed, and strap the monitor back to the shelf or to P2 so it can't slide forward.

P5 Control panel layout

SuperCade_Controls_v3 (27" x 11.3"). Print templates/control-panel-1to1.pdf for the real-size version.



dashed = full deck P5 (29" x 14 3/4"). Template is 27" wide: centre it on the 29" deck. Players stand at the bottom edge.

HOLE SCHEDULE

16	**30 mm**	Action buttons. Use SCREW-IN buttons with a plastic nut - snap-in buttons only grip thin metal/acrylic.
2	**25 mm**	Joystick shafts. The stick's plate bolts on underneath the deck.
2	**24 mm**	Coin buttons (Select on most emulators).
2	**24 mm**	Player Start buttons.
1	**19 mm**	Aux: menu / exit / volume.
12	**5 mm**	Bolt holes: joystick plates (4 per stick - CHECK against your own stick, spacing varies) and deck corners.
4	**4 mm**	Screws around the centre window.
1	**117 x 39 mm**	Centre octagon window: logo plate, USB port, or a small display.

CUTTING ROUND HOLES

Tape the 1:1 template down. Push an awl (or the screw kit's punch tool) through every crosshair. Remove the paper, draw each circle with a compass, then cut with a sharp craft knife in several light passes. Faster and neater: a hole saw or step drill bit of the right size, run by a grown-up.

Cut small first and open up: the button should need a firm push, and the plastic nut does the rest. Keep the deck thin where the controls go (about 10 mm max): screw-in buttons and joysticks are made for thin panels.

Shopping list

Search terms, not specific listings - check sizes and reviews before you buy.

CARDBOARD + FASTENING

☐ **Double-wall corrugated sheets, 48" x 96"**	5-6 sheets. Or appliance boxes (free from appliance/moving stores).
☐ **Kids' cardboard construction kit (screws + nuts)**	Plastic screws, punch/driver tool, safe saw. Get one whose screws come with nuts.
☐ **Extra plastic screws (refill pack)**	Plan on ~150. Each cross panel uses ~10, shelves and deck add more.
☐ **PVA / white glue + hot glue gun**	Glue laminations and flanges; screws clamp while it dries.
☐ **Nylon nuts + bolts, M4/M5 x 25-30 mm, fender washers**	For the joysticks and anything heavy.
☐ **Wide packing tape / gaffer tape**	Edge-protecting the corners and the bottom edge.

CONTROLS + ELECTRONICS

☐ **"Zero delay" 2-player arcade kit (joysticks + 30 mm buttons + USB encoders)**	Pick one with SCREW-IN buttons, 20+ buttons to cover start/coin.
☐ **Screen**	Sized for a 27" monitor. An old 22-27" monitor or a light TV is fine.
☐ **Computer**	Raspberry Pi 4/5 running an arcade front-end, or any old laptop/mini PC.
☐ **Speakers**	Two small speakers that fit a 5 1/4" x 4" oval cut-out.
☐ **Power strip with a switch**	One switch for everything; mount it on the lower shelf.
☐ **Weight for the base**	2 x 10 lb sandbags or bags of rice/gravel on the lower shelf.

TOOLS

☐ **Long metal straightedge (4 ft) + carpenter's square**	For drawing the side panels.
☐ **Box cutter / utility knife + self-healing mat**	Adults do the long cuts. Kids use the kit's safe saw.
☐ **Scoring tool**	The kit's crease roller, a blunt butter knife or an empty ballpoint pen.
☐ **Drill with 1 3/16" (30 mm) and 1" (25 mm) hole saws (optional)**	Clean control-panel holes. Otherwise a compass + craft knife.

SCREW LENGTH - READ BEFORE YOU LAMINATE

Kids' plastic cardboard screws are short: they're made to join two layers of ordinary box cardboard. Before cutting anything, screw a scrap of your flange cardboard to a scrap of your side-panel cardboard, nut on. If it won't reach through a laminated (2-layer) side, keep the sides single-layer and stiffen them with glued ribs instead (page 2), or use nylon nuts and bolts at those joints.

Assembly steps

Dry-fit with screws only, glue once everything lines up.

1. **Test the screws**
   Screw scrap to scrap in the exact layers you'll use (page 6 note).
2. **Draw + cut the sides (P1)**
   Page 2. Cut one, flip it, trace the second. Laminate or rib them. Label the insides 'LEFT' and 'RIGHT'.
3. **Cut + score cross panels (P2-P9)**
   Page 3 sizes, page 4 flanges. Score with the flutes, fold every flange to 90 deg.
4. **Build the control deck (P5 + P12)**
   Glue the doublers under P5. When dry, tape on the 1:1 template and cut all the holes. Mount joysticks and buttons now - much easier on the bench.
5. **Stand the back up (P2)**
   Lay one side flat, inside face up. Place P2 along edge A-B, flanges down against the side. Punch + screw.
6. **Floor + top (P9, P3)**
   Add P9 on H-A and P3 on B-C. You now have a rigid 'C'. Add the second side on top and screw all three.
7. **Stand it up + add the front**
   Stand the cabinet up. Screw in P8, P7, P6 and P5 (control deck, pre-wired). Then the recess, painted black first: P4c floor, P4b screen board, P4a speaker shelf. Last, P4 - the front frame - as one flat piece over the lot.
8. **Shelves (P10, P11)**
   Fold the box beams, screw their flanges in: lower shelf at the marked line, screen shelf so the monitor centres in the window.
9. **Weight + anchor**
   Put 20 lb of sandbags or rice on the lower shelf BEFORE the monitor goes in. Strap the top to a wall stud with an anti-tip furniture strap.
10. **Electronics**
    Computer on the lower shelf, encoder under the deck, monitor on its shelf. Run cables out through the service door.
11. **Glue + finish**
    Once it all works, back out one screw at a time, run glue into the joint, re-screw. Tape the edges, then paint or wrap it. Put Supercade art on the sides!

SAFETY. This is a tall, narrow box with a screen near the top: it WILL tip if it isn't weighted and strapped. Keep it strapped to the wall whenever kids are around it. Box-cutter work is for adults. Keep the power strip off the floor and don't block the monitor's vents.